A SUBSHEAF OF THE CONSTANT SHEAF Z ON AN IRREDUCIBLE SPACE IS A CONSTANT MULTIPLE ON A NONEMPTY OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The termination step is stated by the author as 'This process has to stop after a finite number of steps'; the sentence is reproduced verbatim, and the strictly decreasing integer produced at each step is what makes the descent well founded.

Editorial note. The counted spans carry 1 navigation stub and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-subsheaf-irreducible is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- The constant sheaf Z on an irreducible space has value Z on every nonempty open. Same chapter, the computation recalled from the proof of Lemma lemma-irreducible-constant-cohomology-zero, used at proof lines 3645-3647. Supplies the ambient values against which the sections of the subsheaf are compared.
- Any subgroup of Z is of the form nZ for a nonnegative integer n. Standard group theory, used at proof line 3651. Supplies the generator n of the group of sections.

Source reference. Original lemma 02UW: sources/stacks-cohomology/pinned-original-source.tex, source lines 3462–3462. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be an irreducible topological space. Let $\mathcal{H} \subset \underline{\mathbf{Z}}$ be an abelian subsheaf of the constant sheaf. Then there exists a nonempty open $U \subset X$ such that $\mathcal{H}|_U = \underline{d\mathbf{Z}}_U$ for some $d \in \mathbf{Z}$.*

Proof. Recall that $\underline{\mathbf{Z}}(V) = \mathbf{Z}$ for any nonempty open V of X (see proof of Lemma 02UW). If $\mathcal{H} = 0$, then the lemma holds with $d = 0$. If $\mathcal{H} \neq 0$, then there exists a nonempty open $U \subset X$ such that $\mathcal{H}(U) \neq 0$. Say $\mathcal{H}(U) = n\mathbf{Z}$ for some $n \geq 1$. Hence we see that $\underline{n\mathbf{Z}}_U \subset \mathcal{H}|_U \subset \underline{\mathbf{Z}}_U$. If the first inclusion is strict we can find a nonempty $U' \subset U$ and an integer $1 \leq n' < n$ such that $\underline{n'\mathbf{Z}}_{U'} \subset \mathcal{H}|_{U'} \subset \underline{\mathbf{Z}}_{U'}$. This process has to stop after a finite number of steps, and hence we get the lemma. □

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